

Question 1:

(b) 0/5 points: My answer is as follows: *“False, as sample size grows the central limit theorem implies that the estimated coefficients will converge to their true values if the model is correctly specified. This does not imply that each individual disturbance term becomes normally distributed in fact it implies that the variance of the disturbance terms should converge to zero assuming unbiased estimates.”* I did correctly answer ‘False,’ stating that the CLT is about aggregates/estimators rather than each u_i , and explicitly said that the CLT does not imply normality of individual disturbance terms. Given that the truth value and the core conceptual point about CLT versus individual errors were correct, I am wondering if this might merit some partial credit (e.g., 1–2 points) despite the confusion in the rest of my explanation.

(d) 0/5 points: My answer is as follows: *“True, The Ramsey RESET test uses the values of the auxiliary regression on y to test for the presence of issues that may bias the estimates. These include the presence of OVB, and misspecification of the shape of heteroskedasticity. The RESET test does not say what exactly the problem is, just that there is a problem present using squared values from the auxiliary regression.”* I see now that I incorrectly claimed the Ramsey RESET test can detect misspecification in the shape of heteroskedasticity, which I recognize is a property more squarely addressed by White or Breusch–Pagan. I feel like my answer correctly described the RESET test as focusing on general misspecification through auxiliary regressions with nonlinear terms, and emphasizes its role in identifying problems like omitted nonlinear variables or functional form issues. The error in my answer came from conflating broader diagnostic tests for model issues, but I think the majority of my response aligned with the correct function and purpose of the RESET test. Given this, I am wondering if you would consider an additional 1-2 points for the majority of my response aligning with the answer key.

Question 2:

(a) 1/5 points: My answer is as follows: *“The output from this model suggests that an additional bedroom increases the sales price of a house by \$26,476.185, which according to the bedrooms² coefficient b_2 , decreases by \$2,571.68 per additional bedroom on average holding all else constant. These results (minus the intercept) are all highly statistically significant.”* Here I did not interpret the question as asking for the marginal effect of an additional bedroom from some reference point being the mean as is explained in the answer key. I was focused on how the marginal effect changes with additional bedrooms and by how much per additional bedroom, rather than reporting the marginal effect from a specific reference point since it was not explicitly stated in the question. I feel like this answer demonstrates that I understand the interpretation, as I explain both the marginal effect and how it changes. I based this largely on the fact that previous practice problems varied in their inclusion of a reference point, sometimes containing multiple like in the second practice exam. Given this, I feel like my answer deserves reevaluation as I don't believe 1/5 points is a fair reflection of my understanding of the material.

(b) ½ points: My answer is as follows: “According to model 2, the marginal effect of an additional bedroom is that 1 additional bedroom increases sale price by $b_1 \times 100\%$ or $0.419 \times 100\% = 41.9\%$ which then according to bedrooms²’s b_2 coefficient is decreasing by $-0.044 \times 100 = -4.4\%$ per additional bedroom. This means that adding bedrooms increases the sale price of the house but at a decreasing rate (all else constant). These findings are highly statistically significant at $p < 0.001$.” Similar to part (a), I believe that I correctly interpreted the coefficients in words while not computing the marginal effect from some reference point. Perhaps for more credit I could have included the equation $ME = b_1 + 2b_2$ but I felt like my answer explained the main thing going on here, being a price increase that is decreasing with each consecutive bedroom. Like in part (b) I feel like my answer might deserve more than 1 point.

(d) 0/5 points: I wrote: “This model’s interpretation also changes a bit since the effect of bedrooms on price is now changing with percentage changes in lot size. All else constant, the interpretation would be that the effect of an additional bedroom increases by $0.056 \times 100\%$ when lotsize increases by 1% starting at $-0.329 \times 100\%$ indicating an additional bedroom decreases house price but the amount by which it decreases changes as lotsize increases. The interaction term indicated that the two move together and there may be tradeoffs in determining the sale price of a house.” Similar to parts (a) and (b) I did not interpret this as requesting a point estimate at the average and instead took the approach of explaining the intuition behind the coefficients and how the observed effects may change. I think that if I were to do it again I would include the $ME = b_1 + b_3 \ln(\text{lotsize})$ formula to provide more clarity, and perhaps clean up my wording a bit. Unless I am super missing something though, I think I demonstrated an understanding of the model and its coefficients, in addition to the marginal effects of adding a bedroom. Given that, I would really appreciate consideration for additional credit.

(e) ½ points: I wrote: “Now, the marginal effect of the number of bedrooms on sale price is moving with the house being in a preferred area and with the house having air conditioning. This basically adds to or takes away from the value of the house, intuitively, since we would expect a house with more stuff to be worth more. To start, we can see that the marginal effect of an additional bedroom when both airco and $\text{prefarea} = 0$ is \$7599.547 in sale price. When we include these interaction terms, we can see that when a house has both airco and is in a preferred area, the effect of an additional bedroom on house price is increased by \$14,851. This is significant at the $p < 0.05$ level and the effect of bedrooms alone is significant at $p < 0.001$. We can see that the marginal effect of bedrooms on price when either airco or prefarea is true while the other is false leads to an effect that is not statistically significant, suggesting that they probably vary together.” In this answer, I think I clearly missed the mark on explaining all of the potential marginal effects and instead focused on trying to demonstrate a strong conceptual understanding of the structure of the model and its marginal effects. I see that my answer was a bit wordy and I again did not provide the formula as was included in the answer key, but I feel like my answer demonstrates an understanding of how interactions change the marginal effects in

this case and the magnitude/direction of those changes while providing grounding points. With this, I would be grateful if you could reconsider the 1/5 and possibly award partial credit.